MATH 345 Skeleton Notes

Unit 3: Beyond linear maps: local linearization and first-order optimization

Section 3.6: Directional derivatives, critical points, and gradient descent

---

© 2024–2026 Jacob Denson, Kaitlyn Phillipson, Sebastien Roch. Except where otherwise noted, this work is licensed under the Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International License.

Directional derivatives

By Linear Approximation, the first-order change in a scalar-valued function near $\mathbf{a}$ is

$$f(\mathbf{a}+\mathbf{h}) - f(\mathbf{a}) \approx \nabla f(\mathbf{a})^T \mathbf{h}.$$

To study change along a unit direction $\mathbf{v}$, restrict the input to the line

$$\mathbf{c}(t) = \mathbf{a} + t\mathbf{v}.$$

The resulting one-variable rate of change is the directional derivative. The chain rule for one-parameter inputs in Chain Rule for One Independent Variable gives the gradient formula below.

Definition

Let f be a scalar-valued function. Then if $\mathbf{v}$ is a unit vector, the directional derivative of f in the direction of the vector $\mathbf{v}$, denoted $D_{\mathbf{v}} f$, is defined as

$$D_{\mathbf{v}} f(\mathbf{a}) = \lim_{h \to 0} \frac{f(\mathbf{a}+h\mathbf{v}) - f(\mathbf{a})}{h},$$

provided the limit exists.

The directional derivative represents the rate of change of the function f at the point $\mathbf{a}$ in the direction of the unit vector $\mathbf{v}$. For a function $f(x, y)$ of two variables, the directional derivative $D_{\mathbf{v}} f(\mathbf{a})$ is the slope of the tangent line to the curve obtained by intersecting the surface $z = f(x, y)$ with the plane containing both the line passing through $\mathbf{a}$ with direction vector $\mathbf{v}$, and the z axis.

Remark

Partial derivatives are a special case of directional derivatives, i.e., we have

$$\frac{\partial f}{\partial x_i} = D_{\mathbf{e}_i} f.$$

Despite partial derivatives being a special case of directional derivatives, in most circumstances we can compute directional derivatives using partial derivatives, or equivalently, the gradient.

Theorem

Suppose f is a function differentiable at a point $\mathbf{a}$. Then for each unit vector $\mathbf{v}$, $D_{\mathbf{v}} f(\mathbf{a}) = \nabla f(\mathbf{a}) \cdot \mathbf{v}$.

Why is this true?

Define the path $\mathbf{c}(t) = \mathbf{a} + t\mathbf{v}$ and the single-variable function $g(t) = f(\mathbf{c}(t)) = f(\mathbf{a} + t\mathbf{v})$. Since $\mathbf{c}(0) = \mathbf{a}$, we have $g(0) = f(\mathbf{a})$. Therefore, the definition of the directional derivative gives

$$\begin{aligned} D_{\mathbf{v}} f(\mathbf{a}) &= \lim_{h \to 0} \frac{f(\mathbf{a} + h\mathbf{v}) - f(\mathbf{a})}{h} \\ &= \lim_{h \to 0} \frac{g(h) - g(0)}{h} \\ &= g'(0). \end{aligned}$$

Write $\mathbf{a} = (a_1, \dots, a_n)$ and $\mathbf{v} = (v_1, \dots, v_n)$. The ith coordinate of $\mathbf{c}(t)$ is $c_i(t) = a_i + tv_i$, so $c_i'(0) = v_i$. Because $g = f \circ \mathbf{c}$, the chain rule (Chain Rule for One Independent Variable) gives

$$\begin{aligned} g'(0) &= \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}(\mathbf{c}(0))c_i'(0) \\ &= \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}(\mathbf{a})v_i \\ &= \nabla f(\mathbf{a}) \cdot \mathbf{v}. \end{aligned}$$

Combining the two calculations proves that $D_{\mathbf{v}} f(\mathbf{a}) = \nabla f(\mathbf{a}) \cdot \mathbf{v}$.

The theorem greatly simplifies the computations of directional derivatives.

Example: Directional derivatives and unit directions (U3-LO8)
Let $f(x, y) = x^2 y + y^2$.

Task
Compute the directional derivative $D_{\mathbf{v}} f(1, 2)$, where $\mathbf{v} = (1/\sqrt{2}, 1/\sqrt{2})$, and $f(x, y) = x^2 y + y^2$.

We begin by computing the gradient of f at $(1, 2)$. We have

$$f_x(x, y) = 2xy \quad \text{and} \quad f_y(x, y) = x^2 + 2y.$$

So $\nabla f(x, y) = (2xy, x^2 + 2y)$, and in particular

$$\nabla f(1,2) = \begin{bmatrix} 2(1)(2) \\ (1)^2 + 2(2) \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}.$$

The given vector $\mathbf{v}$ is already a unit vector because

$$\|\mathbf{v}\| = \sqrt{\left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2} = 1.$$

Therefore, we may apply the directional-derivative formula directly:

$$D_{\mathbf{v}} f(1,2) = \nabla f(1,2) \cdot \mathbf{v} = \begin{bmatrix} 4 \\ 5 \end{bmatrix} \cdot \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} = 9/\sqrt{2}.$$

Task

Find the directional derivative of f at the point $(-1, 3)$ in the direction of the vector $(-1, \sqrt{3})$. That is, compute $D_{\mathbf{v}} f(-1, 3)$, where $\mathbf{v}$ is the unit vector in the direction of $(-1, \sqrt{3})$.

The vector

$$\begin{bmatrix} -1 \\ \sqrt{3} \end{bmatrix}$$

has norm

$$\sqrt{(-1)^2 + (\sqrt{3})^2} = 2.$$

So the unit vector in this direction is

$$\mathbf{v} = \frac{1}{2}\begin{bmatrix} -1 \\ \sqrt{3} \end{bmatrix} = \begin{bmatrix} -1/2 \\ \sqrt{3}/2 \end{bmatrix}.$$

For

$$f(x, y) = x^2 y + y^2,$$

we have

$$\nabla f(x, y) = \begin{bmatrix} 2xy \\ x^2 + 2y \end{bmatrix}.$$

Thus

$$\nabla f(-1, 3) = \begin{bmatrix} -6 \\ 7 \end{bmatrix}.$$

Therefore

$$
\begin{aligned}
D_{\mathbf{v}} f(-1,3) &= \nabla f(-1,3) \cdot \mathbf{v} \\
&= \begin{bmatrix} -6 \\ 7 \end{bmatrix} \cdot \begin{bmatrix} -1/2 \\ \sqrt{3}/2 \end{bmatrix} \\
&= 3 + \frac{7\sqrt{3}}{2} = \frac{6 + 7\sqrt{3}}{2}.
\end{aligned}
$$

Activity: A directional derivative between two points (U3-LO8)

Find the directional derivative of f at $(2, 1)$ in the direction from $(2, 1)$ to $(3, 5)$.

The unit vector pointing in the direction from $(2, 1)$ to $(3, 5)$ is the unit vector pointing in the same direction as the vector $(3 - 2, 5 - 1) = (1, 4)$. This vector has magnitude $\sqrt{1^2 + 4^2} = \sqrt{17}$, so the unit vector pointing in this direction is equal to

$$
\mathbf{v} = \begin{bmatrix} 1/\sqrt{17} \\ 4/\sqrt{17} \end{bmatrix}.
$$

Now

$$\nabla f(x, y) = \begin{bmatrix} 3x^2 y \\ x^3 \end{bmatrix},$$

and in particular,

$$\nabla f(2, 1) = \begin{bmatrix} 3(2)^2(1) \\ (2)^3 \end{bmatrix} = \begin{bmatrix} 12 \\ 8 \end{bmatrix}.$$

So

$$\begin{aligned} D_{\mathbf{v}} f(2, 1) &= \nabla f(2, 1) \cdot \mathbf{v} \\ &= \begin{bmatrix} 12 \\ 8 \end{bmatrix} \cdot \begin{bmatrix} 1/\sqrt{17} \\ 4/\sqrt{17} \end{bmatrix} \\ &= \frac{12 + 32}{\sqrt{17}} = \frac{44}{\sqrt{17}}. \end{aligned}$$

Activity: Directional change toward a specified point (U3-LO8)

Let $f(x, y) = x^2 e^y$. Find the directional derivative of f at $(2, 0)$ in the direction pointing from $(2, 0)$ to $(3, 0)$.

A vector pointing from $(2, 0)$ to $(3, 0)$ is

$$\mathbf{d} = \begin{bmatrix} 3-2 \\ 0-0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Its norm is

$$\|\mathbf{d}\| = \sqrt{1^2 + 0^2} = 1.$$

Thus this vector is already a unit vector. Therefore the required direction is

$$\mathbf{v} = \mathbf{d} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \mathbf{e}_1.$$

Next,

$$\nabla f(x, y) = \begin{bmatrix} 2xe^y \\ x^2 e^y \end{bmatrix}, \qquad \nabla f(2, 0) = \begin{bmatrix} 4 \\ 4 \end{bmatrix}.$$

Thus

$$
\begin{aligned}
D_{\mathbf{v}} f(2,0) &= \nabla f(2,0) \cdot \mathbf{v} \\
&= \begin{bmatrix} 4 \\ 4 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\
&= 4 \cdot 1 + 4 \cdot 0 = 4.
\end{aligned}
$$

Activity: Alignment with the gradient (U3-LO8, U1-LO1)

Suppose

$$\nabla f(\mathbf{a}) = \begin{bmatrix} 3 \\ 4 \end{bmatrix}.$$

Compute $D_{\mathbf{v}} f(\mathbf{a})$ for each unit vector below:

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \qquad \mathbf{v}_3 = \frac{1}{5}\begin{bmatrix} 3 \\ 4 \end{bmatrix}, \qquad \mathbf{v}_4 = -\frac{1}{5}\begin{bmatrix} 3 \\ 4 \end{bmatrix}.$$

Which listed direction gives the greatest increase? Which gives the greatest decrease?

By the theorem,

$$D_{\mathbf{v}_1} f(\mathbf{a}) = \begin{bmatrix} 3 \\ 4 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 3,$$

and

$$D_{\mathbf{v}_2} f(\mathbf{a}) = \begin{bmatrix} 3 \\ 4 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 4.$$

Also,

$$D_{\mathbf{v}_3} f(\mathbf{a}) = \begin{bmatrix} 3 \\ 4 \end{bmatrix} \cdot \frac{1}{5} \begin{bmatrix} 3 \\ 4 \end{bmatrix} = 5,$$

while

$$D_{\mathbf{v}_4} f(\mathbf{a}) = -5.$$

The greatest listed increase occurs in direction $\mathbf{v}_3$, which is aligned with the gradient. The greatest listed decrease occurs in direction $\mathbf{v}_4$, which points opposite the gradient.

Theorem

Suppose that a function f is differentiable at a point $\mathbf{a}$.

- If $\nabla f(\mathbf{a}) = 0$, then $D_{\mathbf{v}} f(\mathbf{a}) = 0$ for any unit vector $\mathbf{v}$.
- If $\nabla f(\mathbf{a}) \neq \mathbf{0}$, then $D_{\mathbf{v}} f(\mathbf{a})$ is maximized, among unit vectors $\mathbf{v}$, when

$$\mathbf{v} = \frac{\nabla f(\mathbf{a})}{\|\nabla f(\mathbf{a})\|}.$$

It is minimized when

$$\mathbf{v} = -\frac{\nabla f(\mathbf{a})}{\|\nabla f(\mathbf{a})\|}.$$

The maximum value is $\|\nabla f(\mathbf{a})\|$, and the minimum value is $-\|\nabla f(\mathbf{a})\|$.

Why is this true?

If $\nabla f(\mathbf{a}) = \mathbf{0}$, then the theorem gives

$$D_{\mathbf{v}} f(\mathbf{a}) = 0$$

for every unit vector $\mathbf{v}$.

Now suppose $\nabla f(\mathbf{a}) \neq \mathbf{0}$, and let θ be the angle between $\nabla f(\mathbf{a})$ and a unit vector $\mathbf{v}$. By the theorem and Angle between vectors,

$$D_{\mathbf{v}} f(\mathbf{a}) = \nabla f(\mathbf{a}) \cdot \mathbf{v} = \|\nabla f(\mathbf{a})\| \cos\theta.$$

The largest possible value of $\cos\theta$ is 1, attained when $\mathbf{v}$ points in the direction of the gradient. The smallest possible value is -1, attained when $\mathbf{v}$ points in the opposite direction. Therefore the maximum and minimum directional derivatives are

$$\|\nabla f(\mathbf{a})\| \quad \text{and} \quad -\|\nabla f(\mathbf{a})\|,$$

attained at

$$\mathbf{v} = \frac{\nabla f(\mathbf{a})}{\|\nabla f(\mathbf{a})\|} \quad \text{and} \quad \mathbf{v} = -\frac{\nabla f(\mathbf{a})}{\|\nabla f(\mathbf{a})\|},$$

respectively.

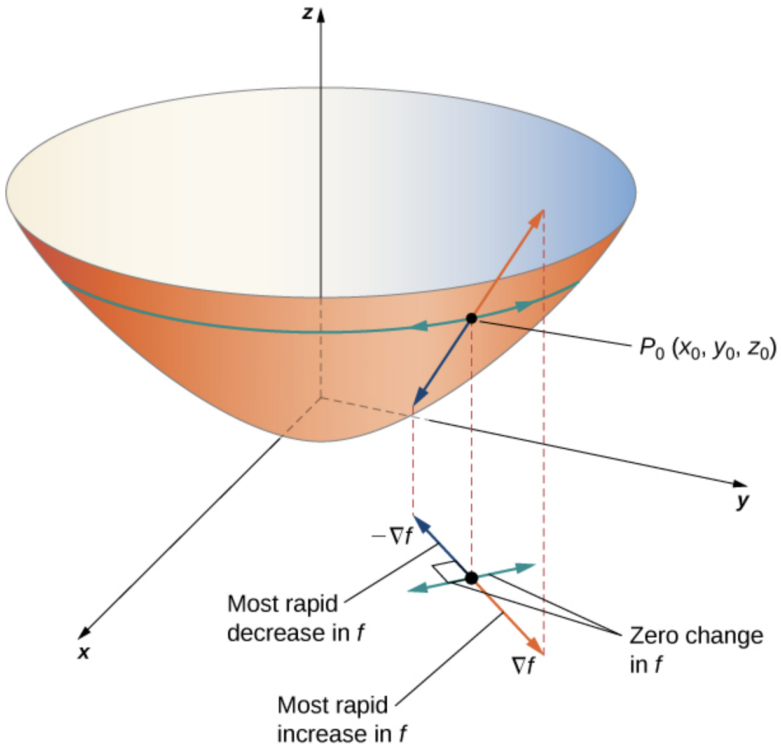


Figure. Gradient directions at a point on a surface.

Activity: Finding the direction of greatest increase (U3-LO8)

Find the direction for which the directional derivative of $f(x, y) = 3x^2 - 4xy + 2y^2$ at $(-2, 3)$ is a maximum. What is the maximum value of the directional derivative?

First, we calculate that

$$\nabla f(x, y) = \begin{bmatrix} 6x - 4y \\ -4x + 4y \end{bmatrix}.$$

So

$$\nabla f(-2,3) = \begin{bmatrix} 6(-2) - 4(3) \\ -4(-2) + 4(3) \end{bmatrix} = \begin{bmatrix} -24 \\ 20 \end{bmatrix}.$$

the theorem tells us that the directional derivative $D_{\mathbf{v}} f$ is maximized when $\mathbf{v}$ is the unit vector pointing in the direction of $(-24, 20)$. Since $(-24, 20)$ has magnitude $\sqrt{(-24)^2 + (20)^2} = \sqrt{976}$, we see the direction which maximizes the directional derivative is

$$\mathbf{v} = \begin{bmatrix} -24/\sqrt{976} \\ 20/\sqrt{976} \end{bmatrix},$$

and the value of the directional derivative in this direction is equal to the magnitude of the gradient, which is

$$\sqrt{976}.$$

Activity: A directional derivative in three variables (U3-LO8)

Calculate $D_{\mathbf{u}} f(1, -2, 3)$ in the direction of $\mathbf{v} = -\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$ for the function

Activity: A directional derivative in three variables (U3-LO8) (continued)

$$f(x, y, z) = 5x^2 - 2xy + y^2 - 4yz + z^2 + 3xz.$$

We compute that

$$\|\mathbf{v}\| = \sqrt{(-1)^2 + (2)^2 + (2)^2} = \sqrt{9} = 3$$

So the unit vector in the direction of $\mathbf{v}$ is the vector

$$\mathbf{u} = \frac{\mathbf{v}}{\|\mathbf{v}\|} = -\frac{1}{3}\mathbf{i} + \frac{2}{3}\mathbf{j} + \frac{2}{3}\mathbf{k}$$

Using the formula for the directional derivative, we find that

$$
\begin{aligned}
D_{\mathbf{u}} f(x,y,z) &= \nabla f(x,y,z) \cdot \mathbf{u} \\
&= \begin{bmatrix} 10x - 2y + 3z \\ -2x + 2y - 4z \\ -4y + 2z + 3x \end{bmatrix} \cdot \mathbf{u} \\
&= -\frac{10x - 2y + 3z}{3} \\
&\quad + 2\frac{-2x + 2y - 4z}{3} \\
&\quad + 2\frac{-4y + 2z + 3x}{3}.
\end{aligned}
$$

At the point $(1, -2, 3)$, we evaluate that

$$
\begin{aligned}
D_{\mathbf{u}} f(1,-2,3) &= (23)\left(-\frac{1}{3}\right) \\
&\quad + (-18)\left(\frac{2}{3}\right) + (17)\left(\frac{2}{3}\right) \\
&= \frac{-25}{3}
\end{aligned}
$$

In Orthogonal projection, the same dot product will be interpreted as a signed projection coefficient of the gradient onto the chosen direction.

Optimization problems and critical points

Directional derivatives describe first-order change along a chosen direction. We now use them to identify candidates for local extrema and to motivate an iterative method for decreasing a function.
In this unit, we apply multivariate calculus to solve *optimization problems*.

Definition: Optimization problems

Let $f : \mathbb{R}^n \to \mathbb{R}$ be a scalar-valued function, and consider a set $U \subseteq \mathbb{R}^n$. Then we can consider two optimization problems, in which we call f the objective function of the problem, and U the feasible region of the optimization problem:

- The minimization problem of f on U is the problem of finding $\mathbf{x}^* \in U$ such that the value $f(\mathbf{x}^*)$ is *smallest* among all elements of U (so the function is *minimized* on U by $\mathbf{x}^*$).
- The maximization problem of f on U is the problem of finding $\mathbf{x}^* \in U$ such that the value $f(\mathbf{x}^*)$ is *largest* among all elements of U (so the function is *maximized* on U by $\mathbf{x}^*$).

Definition: Minima and maxima

Suppose f is a scalar-valued function. A point $\mathbf{a}$ in the domain of f is a

- local minimum if there is a ball centered at $\mathbf{a}$ such that $f(\mathbf{a}) \leq f(\mathbf{x})$ whenever $\mathbf{x}$ is in that ball and also in the domain of f; in other words, whenever $\mathbf{x}$ is close enough to $\mathbf{a}$ and lies in the domain of f. In this case, we say $f(\mathbf{a})$ is a local minimum value for the function f.

- local maximum if there is a ball centered at $\mathbf{a}$ such that $f(\mathbf{a}) \geq f(\mathbf{x})$ whenever $\mathbf{x}$ is in that ball and also in the domain of f; in other words, whenever $\mathbf{x}$ is close enough to $\mathbf{a}$ and lies in the domain of f. In this case, we say $f(\mathbf{a})$ is a local maximum value for the function f.
- global minimum if $f(\mathbf{a}) \leq f(\mathbf{x})$ for all $\mathbf{x}$ in the domain of f. In this case, $f(\mathbf{a})$ is the global minimum value, or absolute minimum value, of the function f.
- global maximum if $f(\mathbf{a}) \geq f(\mathbf{x})$ for all $\mathbf{x}$ in the domain of f. In this case, $f(\mathbf{a})$ is the global maximum value, or absolute maximum value, of the function f.

A local minimum or local maximum is also called a local extremum.

Remark

The graph of the function $-f$ is obtained by reflecting the graph of the function f in the output axis. This operation turns local and global maxima into local and global minima, and vice versa. So methods for finding minima for a general function often give methods for finding maxima, which makes the two problems often equivalent.

Definition: Critical points

Suppose f is a scalar-valued function that is differentiable at a point $\mathbf{a}$.

Then we say $\mathbf{a}$ is a **critical point** if $\nabla f(\mathbf{a}) = \mathbf{0}$.

For a scalar-valued function of two variables, for example, $f(x, y)$, a critical point is a solution to the two equations $f_x(x, y) = 0$ and $f_y(x, y) = 0$. Just as the derivative of a scalar-valued function of a single variable helps identify local minima and maxima, partial derivatives provide information about possible local minima/maxima.

Theorem

Suppose f is differentiable at an interior point $\mathbf{a}$ of its domain. If $\mathbf{a}$ is a local extremum of f, then

$$\nabla f(\mathbf{a}) = \mathbf{0}.$$

Why is this true?

Let $\mathbf{v}$ be a unit vector, and define

$$g(t) = f(\mathbf{a} + t\mathbf{v}).$$

Because $\mathbf{a}$ is an interior local extremum of f, the function g has a local extremum at $t = 0$. Therefore

$$g'(0) = 0.$$

By the theorem,

$$g'(0) = D_{\mathbf{v}} f(\mathbf{a}) = \nabla f(\mathbf{a}) \cdot \mathbf{v}.$$

This holds for every unit vector $\mathbf{v}$. If $\nabla f(\mathbf{a}) \neq \mathbf{0}$, choose

$$\mathbf{v} = \frac{\nabla f(\mathbf{a})}{\|\nabla f(\mathbf{a})\|}.$$

Then

$$\nabla f(\mathbf{a}) \cdot \mathbf{v} = \|\nabla f(\mathbf{a})\| > 0,$$

a contradiction. Hence $\nabla f(\mathbf{a}) = \mathbf{0}$.

Activity: Critical points of a quadratic function (U3-LO9)

Find the critical points of the function

$$f(x, y) = x^2 + y^2 - 2x + 4y.$$

To find the critical points, we calculate the partial derivatives and set them equal to zero. We find that

$$f_x(x, y) = 2x - 2 \quad \text{and} \quad f_y(x, y) = 2y + 4.$$

The critical points are the pairs (x, y) that simultaneously solve the equations $2x - 2 = 0$ and $2y + 4 = 0$. So the only critical point is $(1, -2)$.

Activity: Critical points of a difference of squares (U3-LO9)

Find the critical points of $f(x, y) = x^2 - y^2$.

We compute that $f_x(x, y) = 2x$ and $f_y(x, y) = -2y$. The only (x, y) such that $f_x(x, y) = 2x = 0$ and $f_y(x, y) = -2y = 0$ is $(0, 0)$. This is the only critical point.

Activity: Critical points via row reduction (U3-LO9, U2-LO2)

Find all critical points of

$$f(x, y, z) = x^2 + xy + yz + z^2 - x - 2y - z.$$

Write the critical-point equations as an augmented matrix, use Gaussian elimination to obtain row echelon form, and finish by back-substitution.

The polynomial f is differentiable everywhere, and its gradient is

$$\nabla f(x, y, z) = \begin{bmatrix} 2x + y - 1 \\ x + z - 2 \\ y + 2z - 1 \end{bmatrix}.$$

Thus its critical points are exactly the solutions of

$$\begin{cases} 2x + y = 1, \\ x + z = 2, \\ y + 2z = 1. \end{cases}$$

Following The Gaussian Elimination Algorithm, we first swap rows to obtain a leading pivot of 1. We then eliminate below each pivot and normalize the final pivot:

$$\left[\begin{array}{ccc|c} 2 & 1 & 0 & 1 \\ 1 & 0 & 1 & 2 \\ 0 & 1 & 2 & 1 \end{array}\right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 2 & 1 & 0 & 1 \\ 0 & 1 & 2 & 1 \end{array}\right]$$

$$\xrightarrow{R_2 \to R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & -2 & -3 \\ 0 & 1 & 2 & 1 \end{array}\right]$$

$$\xrightarrow{R_3 \to R_3 - R_2} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & -2 & -3 \\ 0 & 0 & 4 & 4 \end{array}\right]$$

$$\xrightarrow{R_3 \to \frac{1}{4}R_3} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & -2 & -3 \\ 0 & 0 & 1 & 1 \end{array}\right].$$

The last row gives $z = 1$. Back-substitution into the second row, $y - 2z = -3$, and then the first row, $x + z = 2$, gives

$$y = -3 + 2z = -3 + 2(1) = -1,$$
$$x = 2 - z = 2 - 1 = 1.$$

There is a pivot in every variable column and no inconsistent row, so the system has a unique solution. Therefore the only critical point of f is $(1, -1, 1)$.

Warning: Critical points are candidates

The equation $\nabla f = 0$ finds candidates for local extrema. It does not by itself find boundary extrema, constrained extrema, or absolute extrema. Always combine critical-point information with the domain and boundary conditions.

Gradient descent

Example: Critical-point equations in three variables (U3-LO9)

Find the system of equations satisfied by the critical points of the function

$$f(x, y, z) = x^2 e^{yz} + y \sin(xz) + z \cos(xy).$$

We calculate that

$$f_x(x, y, z) = 2xe^{yz} + yz\cos(xz) - yz\sin(xy),$$

that

$$f_y(x, y, z) = x^2ze^{yz} + \sin(xz) - xz\sin(xy)$$

and that

$$f_z(x, y, z) = x^2ye^{yz} + xy\cos(xz) + \cos(xy).$$

So the critical points of f are the solutions to the system of equations

$$\begin{cases} 2xe^{yz} + yz\cos(xz) - yz\sin(xy) = 0, \\ x^2ze^{yz} + \sin(xz) - xz\sin(xy) = 0, \\ x^2ye^{yz} + xy\cos(xz) + \cos(xy) = 0. \end{cases}$$

Using the calculus we have developed, it is usually straightforward to write the equations that critical points must satisfy. Critical-point equa-

tions in three variables (U3-LO9) also shows the problem: those equations may be nonlinear and difficult or impossible to solve exactly. This is different from the linear systems in Unit 2, where Gaussian elimination gave a systematic exact method.

When the critical-point equations cannot be solved directly, one often uses a numerical method. By the theorem, the negative gradient gives the direction of steepest local decrease.

At a current point $\mathbf{x}_k$, choose

$$\mathbf{h} = -\alpha \nabla f(\mathbf{x}_k), \qquad \alpha > 0.$$

By Linear Approximation,

$$\begin{aligned} f(\mathbf{x}_k + \mathbf{h}) - f(\mathbf{x}_k) &\approx \nabla f(\mathbf{x}_k)^T \mathbf{h} \\ &= -\alpha \|\nabla f(\mathbf{x}_k)\|^2. \end{aligned}$$

Unless the gradient is zero, the local model predicts a decrease. This is a local prediction, not a convergence theorem. It motivates the gradient-descent update

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha \nabla f(\mathbf{x}_k).$$

The gradient-descent algorithm repeats this local step.

1. Start at an initial point $\mathbf{x}_0$.
2. Compute the gradient $\nabla f(\mathbf{x}_0)$.
3. Move in the direction of the negative gradient: set $\mathbf{x}_1 = \mathbf{x}_0 - \alpha_0 \nabla f(\mathbf{x}_0)$, where $\alpha_0 > 0$ is a step size (or learning rate) parameter.
4. Repeat steps 2-3 with step sizes $\alpha_1, \alpha_2, \ldots$ to successively define points $\mathbf{x}_2 = \mathbf{x}_1 - \alpha_1 \nabla f(\mathbf{x}_1)$, $\mathbf{x}_3 = \mathbf{x}_2 - \alpha_2 \nabla f(\mathbf{x}_2)$, and so on, with the hope that these points converge to a local minimum.

In data and modeling problems, an objective function being minimized is often called a loss.

Intuitively, gradient descent works because each successive value $\mathbf{x}_k$ is obtained by shifting the previous value $\mathbf{x}_{k-1}$ in the direction in which the function f decreases the fastest. Gradient descent may converge to a local minimum under additional assumptions on the function and the learning rates. The examples below illustrate possible behavior; they do not prove convergence.

Activity: Reading one gradient descent update (U3-LO8, U3-LO9)

Assume `grad_f(x)` computes $\nabla f(\mathbf{x})$. Consider the line of code

```
x = x - alpha * grad_f(x)
```

1. What mathematical update rule does this line represent?
2. Which quantity is the learning rate?
3. Why must `grad_f(x)` have the same shape as `x`?
4. What would change if the minus sign were a plus sign?

The line represents the update

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha \nabla f(\mathbf{x}_k).$$

The learning rate is `alpha`. The vector `grad_f(x)` must have the same shape as `x`, because the update subtracts one vector from

another. If the minus sign were a plus sign, the update would move in the direction of steepest increase rather than steepest decrease.

Activity: Three gradient-descent steps (U3-LO9)

Find an approximate minimum of the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by

$$f(x, y) = x^2 - 3xy + 3y^2 + 5y + 2x$$

using gradient descent. Use the starting point $\mathbf{x}_0 = (0, 0)$, a constant step size of 0.1, and stop after computing $\mathbf{x}_3$. You may use a calculator to perform these computations.

First, we calculate the gradient:

$$\nabla f(x, y) = \begin{bmatrix} 2x - 3y + 2 \\ -3x + 6y + 5 \end{bmatrix}$$

Starting at $\mathbf{x}_0 = (0, 0)$, we successively compute that

$$\nabla f(0, 0) = \begin{bmatrix} 2 \\ 5 \end{bmatrix},$$

so

$$\mathbf{x}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} - 0.1 \begin{bmatrix} 2 \\ 5 \end{bmatrix} = \begin{bmatrix} -0.2 \\ -0.5 \end{bmatrix}.$$

We then compute that

$$\begin{aligned} &\nabla f(-0.2, -0.5) \\ &= \begin{bmatrix} -0.4 - (-1.5) + 2 \\ -3(-0.2) + 6(-0.5) + 5 \end{bmatrix} \\ &= \begin{bmatrix} 3.1 \\ 2.6 \end{bmatrix}, \end{aligned}$$

so

$$\mathbf{x}_2 = \begin{bmatrix} -0.2 \\ -0.5 \end{bmatrix} - 0.1 \begin{bmatrix} 3.1 \\ 2.6 \end{bmatrix} = \begin{bmatrix} -0.51 \\ -0.76 \end{bmatrix}.$$

Finally, we compute that

$$\begin{aligned}\nabla f(-0.51, -0.76) &= \begin{bmatrix} 2(-0.51) - 3(-0.76) + 2 \\ -3(-0.51) + 6(-0.76) + 5 \end{bmatrix} \\ &= \begin{bmatrix} 3.26 \\ 1.97 \end{bmatrix},\end{aligned}$$

so

$$\mathbf{x}_3 = \begin{bmatrix} -0.51 \\ -0.76 \end{bmatrix} - 0.1 \begin{bmatrix} 3.26 \\ 1.97 \end{bmatrix} = \begin{bmatrix} -0.836 \\ -0.957 \end{bmatrix}.$$

In Revisiting the gradient-descent quadratic (U5-LO5, U3-LO9), we compute exactly that the unique critical point is

$$\left(-9, -\frac{16}{3}\right)$$

and classify it after developing second-order tools. Thus $\mathbf{x}_3$ is not yet a good approximation. If we continue the iteration, the points $\mathbf{x}_k$ for, say, $k \geq 1000$ are very close to that critical point.

Activity: From a symbolic gradient to a descent step (U3-LO9)

Use SymPy to check the first step of the preceding calculation for $f(x, y) = x^2 - 3xy + 3y^2 + 5y + 2x$. We construct a gradient as in Building a gradient and Hessian in SymPy (U3-LO3), evaluate it at the starting point, and use the vector update from Reading one gradient descent update (U3-LO8, U3-LO9). The learning rate `sp.Rational(1, 10)` represents 0.1 exactly.

```
import sympy as sp

x, y = sp.symbols("x y")
f = x**2 - 3*x*y + 3*y**2 + 5*y + 2*x
grad_f = sp.Matrix([sp.diff(f, x), sp.diff(f, y)])

p0 = sp.Matrix([0, 0])
g0 = grad_f.subs({x: p0[0], y: p0[1]})
alpha = sp.Rational(1, 10)
p1 = p0 - alpha*g0
f0 = f.subs({x: p0[0], y: p0[1]})
f1 = f.subs({x: p1[0], y: p1[1]})

print(grad_f)
print(g0)
print(p1)
print(f0, f1)
print((g0.T * (p1 - p0))[0], f1 - f0)
```

Output:

```
Matrix([[2*x - 3*y + 2], [-3*x + 6*y + 5]])
Matrix([[2], [5]])
Matrix([[-1/5], [-1/2]])
0 -241/100
-29/10 -241/100
```

In the last line, `g0.T * (p1 - p0)` is a 1×1 matrix. The index `[0]`

Activity: From a symbolic gradient to a descent step (U3-LO9) (continued)

extracts its scalar entry, the local linear prediction for the change in f.

1. What is the difference between `grad_f` and `g0`? What shapes must `p0`, `g0`, and `p1` have for the update to make sense?
2. Write the numerical vector update that produces `p1`. Did the function value decrease? Compare the predicted change with the actual change, using the scalar linear approximation from Linear Approximation.
3. Predict `grad_f.subs({x: p1[0], y: p1[1]})`. Is `p1` a critical point? Explain which symbolic expression can be reused at the next step and which evaluated vector must be recomputed.
4. If `alpha` is changed to `1` and the subsequent lines are rerun, what are `p1` and `f1`? What does this comparison show about the learning rate?

1. The matrix `grad_f` contains expressions for the two partial derivatives as functions of x, y; `g0` is their value at $(0, 0)$. Each point and evaluated gradient is a 2×1 column, so the scalar multiple `alpha*g0` can be subtracted from `p0`.
2. The update is

$$\mathbf{p}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} - \frac{1}{10}\begin{bmatrix} 2 \\ 5 \end{bmatrix} = \begin{bmatrix} -1/5 \\ -1/2 \end{bmatrix}.$$

The function value decreases from 0 to $-241/100 = -2.41$. The local linear approximation predicts

$$\nabla f(\mathbf{p}_0)^T(\mathbf{p}_1 - \mathbf{p}_0) = -\frac{1}{10}(2^2 + 5^2) = -\frac{29}{10}.$$

The actual change is $-241/100$. Both are negative, but the actual decrease is smaller than predicted because the linear approximation omits nonlinear terms.

3. Evaluating at the new point gives

$$\nabla f(-1/5, -1/2) = \begin{bmatrix} 31/10 \\ 13/5 \end{bmatrix},$$

so $\mathbf{p}_1$ is not a critical point. The symbolic expression `grad_f` can be reused without differentiating again. Its value must be recomputed at the new point; reusing the old

vector `g0` would not implement gradient descent at that point.

4. With $\alpha = 1$, the new point is $(-2, -5)$ and $f(-2, -5) = 20$, an increase from 0. The local model predicts a change of -29, but this larger step goes too far for that prediction to ensure a decrease. A negative-gradient direction does not guarantee a decrease for every positive learning rate.

Remark

The iteration is cumbersome by hand but easy to repeat on a computer when the gradient is available.

First gradient descent steps

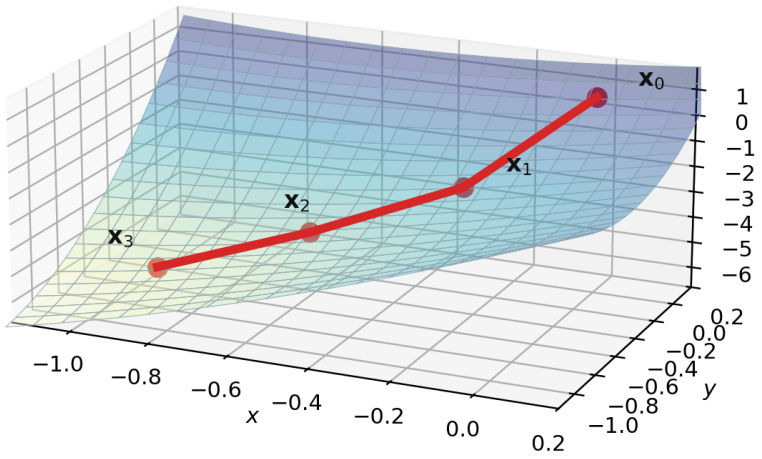


Figure. Surface plot showing the first gradient descent steps from the origin.

Activity: Improving a model's predictions with one step (U3-LO8, U3-LO9)

Return to the three observations from From predictions to a scalar error function (U3-LO1, U2-LO1) and the loss

$$\mathscr{L}(w_1, w_2) = (w_1 - 1)^2 + (w_2 - 2)^2 + (w_1 + w_2 - 2)^2.$$

At the starting weights $\mathbf{w}_0 = (1, 1)$, the loss is 1 and the gradient is $(0, -2)$. In this model, a training step adjusts the weights while the features X and observations $\mathbf{y}$ remain fixed.

1. Find the unit direction of greatest decrease at $\mathbf{w}_0$ and its directional derivative. Take one gradient-descent step with learning rate $\alpha = 1/10$.
2. Compute the new prediction vector, residual, and loss. Compare the actual change with the first-order prediction $-\alpha\|\nabla\mathscr{L}(\mathbf{w}_0)\|^2$.
3. Evaluate the gradient at the new weights. Has the step reached a critical point?

Activity: Improving a model's predictions with one step (U3-LO8, U3-LO9) (continued)

4. Starting again at $\mathbf{w}_0$, what happens if $\alpha = 1$? Explain why the same descent direction can give an increase in loss.

1. The negative gradient is $(0, 2)$, so its unit direction is $(0, 1)$. The directional derivative is $(0, -2) \cdot (0, 1) = -2$. Gradient descent uses the unnormalized gradient:

$$\mathbf{w}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \frac{1}{10} \begin{bmatrix} 0 \\ -2 \end{bmatrix} = \begin{bmatrix} 1 \\ 6/5 \end{bmatrix}.$$

2. The new predictions and residuals are

$$X\mathbf{w}_1 = \begin{bmatrix} 1 \\ 6/5 \\ 11/5 \end{bmatrix}, \qquad R(\mathbf{w}_1) = \begin{bmatrix} 0 \\ -4/5 \\ 1/5 \end{bmatrix}.$$

The loss is $16/25 + 1/25 = 17/25 = 0.68$, down from 1. The predicted change is $-(1/10)(4) = -2/5$, while the actual change is $-8/25$. Their difference, $2/25$,

is the squared length of $X(\mathbf{w}_1 - \mathbf{w}_0) = (0, 1/5, 1/5)$, as in Predicting a change in fitting error (U3-LO4).

3. The gradient formula from Differentiating a loss through its predictions (U3-LO6) gives

$$\nabla \mathscr{L}(\mathbf{w}_1) = 2X^T \begin{bmatrix} 0 \\ -4/5 \\ 1/5 \end{bmatrix} = \begin{bmatrix} 2/5 \\ -6/5 \end{bmatrix}.$$

   It is nonzero, so the new weights are not a critical point. The step improves the fit without finishing the minimization.

4. With $\alpha = 1$, the new weights are $(1, 3)$, the predictions are $(1, 3, 4)$, and the residual is $(0, 1, 2)$. The loss becomes 5, which is larger than the starting value. The negative gradient specifies a direction of local decrease; the learning rate determines how far we travel. A step that is too large can increase

the loss, just as in From a symbolic gradient to a descent step (U3-LO9).

Activity: Redundant features give a flat direction of the loss (U3-LO6, U2-LO6)

Recall Unit 2's redundant-feature example in Two coefficient choices, same predictions (U2-LO1, U2-LO6). Add a third feature to our data by summing the original two feature columns:

$$\widetilde{X} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix}, \qquad \mathbf{y} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}.$$

There are now three weights $\theta = (\theta_1, \theta_2, \theta_3)$, with loss

$$\widetilde{\mathscr{L}}(\theta) = \|\widetilde{X}\theta - \mathbf{y}\|^2.$$

The predictions depend only on the two combinations $w_1 = \theta_1 + \theta_3$ and $w_2 = \theta_2 + \theta_3$, so $\widetilde{\mathscr{L}}(\theta) = \mathscr{L}(\theta_1 + \theta_3, \theta_2 + \theta_3)$.

1. Verify that $\mathbf{z} = (-1, -1, 1)$ is in the null space of $\widetilde{X}$.
2. Show that $\widetilde{\mathscr{L}}(\theta + t\mathbf{z}) = \widetilde{\mathscr{L}}(\theta)$ for every real t. Evaluate this constant loss along the line through $\theta_0 = (1, 1, 0)$.
3. What is $\frac{d}{dt}\widetilde{\mathscr{L}}(\theta + t\mathbf{z})$? If a minimizing weight vector is found, can it be the unique minimizing weight vector? Compare this exact invariance with the local Jacobian example in Interpreting a Jacobian's null space in SymPy (U3-LO5, U2-LO3).

1. Direct multiplication verifies the given direction:

$$\widetilde{X}\mathbf{z} = \begin{bmatrix} -1+1 \\ -1+1 \\ -1-1+2 \end{bmatrix} = \mathbf{0}.$$

   Decreasing each original weight by t and increasing the redundant-feature weight by t leaves both effective weights unchanged.

2. Unit 2's linearity identity gives

$$\widetilde{X}(\theta + t\mathbf{z}) - \mathbf{y} = \widetilde{X}\theta - \mathbf{y} + t\widetilde{X}\mathbf{z} = \widetilde{X}\theta - \mathbf{y}.$$

   Taking squared lengths proves equality of the losses for every t. At θ_0, the predictions are $(1, 1, 2)$ and the loss is 1. Every point $(1-t, 1-t, t)$ gives these same predictions and the same loss.

3. The loss along the line is constant, so its derivative is zero for every t. By the chain

rule, $\nabla\widetilde{\mathscr{L}}(\theta)^T\mathbf{z} = 0$. If θ_* minimizes the loss, all distinct vectors $\theta_* + t\mathbf{z}$ minimize it as well. The data therefore cannot identify these three weights uniquely.

This flat direction is exact because the prediction map is linear and $\widetilde{X}\mathbf{z} = \mathbf{0}$. For a general nonlinear map, a vector in the null space of its Jacobian at one point need only give zero first-order change; higher-order terms may still change the output. Also, equal scalar losses alone need not mean equal predictions, as the two choices in From predictions to a scalar error function (U3-LO1, U2-LO1) showed.

These examples use derivatives to improve a fit and interpret directions that leave it unchanged. Unit 4 develops the general least-squares method for finding a best-fitting prediction vector and its geometric interpretation.